

Exercise of Supervised Learning: Gaussian Processes Part 2

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Exercise 1: Gaussian Posterior Process

Assume your data follows the following law:

$$\mathbf{y} = \mathbf{f} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}),$$

with $\mathbf{f} = f(\mathbf{x}) \in \mathbb{R}^n$ being a realization of a Gaussian process (GP), for which we a priori assume

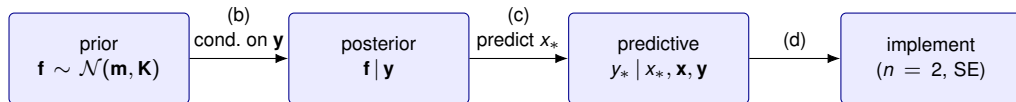
$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')).$$

$\mathbf{x}$ is a single (1-D) feature, observed at n points.

(a) Derive / define the prior distribution of $\mathbf{f}$.

Exercise 1: The GP-Regression Pipeline

Parts (a)–(d) walk from the prior to a working implementation. Each arrow is one operation on a Gaussian:



Key tool throughout: a **Gaussian stays Gaussian** under multiplication (likelihood $\times$ prior) and under conditioning. So every box above is again a Gaussian — we only need to track its mean and covariance.

Exercise 1: Three Gaussians — Don't Confuse Them

The same setup produces three different distributions. Be precise about which one you mean:

Distribution	Symbol	What it is
Posterior	$\mathbf{f} \mid \mathbf{y} \sim \mathcal{N}(\mathbf{f}_{\text{post}}, \mathbf{K}_{\text{post}})$	latent $\mathbf{f}$ <i>after</i> seeing $\mathbf{y}$ (part (b))
Predictive	$y_* \mid x_*, \mathbf{x}, \mathbf{y}$	a <i>new noisy</i> target at x_* (part (c))
Marginal	$\mathbf{y} \sim \mathcal{N}(\mathbf{m}, \mathbf{K} + \sigma^2 \mathbf{I})$	data likelihood; used to <i>train</i> (part (d))

1 (a): Prior Distribution of $\mathbf{f}$

- ▶ $\mathbf{f} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$.
- ▶ $\mathbf{m} = m(\mathbf{x})$.
- ▶ $\mathbf{K}_{ij} = k(x^{(i)}, x^{(j)})$.
- ▶ **NB (finite-dim. property):** for *any* finite set of input points, the corresponding process values are jointly Gaussian $\Rightarrow$ the prior over $\mathbf{f}$ is the finite-dimensional Gaussian above. $\square$

Roadmap (Ex. 1)

▶ (a) Prior

$$\mathbf{f} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

(b) Posterior

$\mathbf{f} | \mathbf{y}$: complete sq.

(c) Predictive

$y_* | \dots$: condition

(d) Implement

2×2 by hand + code

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Exercise 1 (b)

(b) Derive the posterior distribution of $\mathbf{f}|\mathbf{y}$.

1 (b): Think Before Solving

Target: get the posterior $\mathbf{f} \mid \mathbf{y}$ from likelihood and prior — without computing any normalizing integral.

- ▶ **Both factors are Gaussian.** Bayes gives $p(\mathbf{f} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{f}) p(\mathbf{f})$. A product of two Gaussian densities (in $\mathbf{f}$) is again Gaussian — so what kind of function of $\mathbf{f}$ sits in the exponent?
- ▶ **Complete the square.** The exponent is a quadratic $-\frac{1}{2} \mathbf{f}^\top A \mathbf{f} + \mathbf{f}^\top b$. Which matrix plays the role of A , and which vector plays b ?
- ▶ **Read off the parameters.** Once written as $-\frac{1}{2} (\mathbf{f} - \mu)^\top A (\mathbf{f} - \mu)$, the mean and covariance fall out directly. What are they?

Gaussian $\times$ Gaussian $\rightarrow$ complete the square $\rightarrow$ read off $(\mathbf{f}_{\text{post}}, K_{\text{post}})$

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1 (b): Likelihood and Prior

The posterior can be derived from the likelihood and prior using Bayes rule:

$$\mathbf{y} = \mathbf{f} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

The likelihood is:

$$p(\mathbf{y}|\mathbf{f}) \propto \exp \left(-\frac{1}{2}(\mathbf{y} - \mathbf{f})^\top (\sigma^2 \mathbf{I})^{-1} (\mathbf{y} - \mathbf{f}) \right).$$

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1 (b): Complete the Square

$$p(\mathbf{f}|\mathbf{y}) \propto \exp \left(-\frac{1}{2} \mathbf{f}^\top \mathbf{K}_{\text{post}}^{-1} \mathbf{f} + \mathbf{f}^\top \tilde{\mathbf{f}} \right)$$

Completing the square (symmetric $\mathbf{A}$):

- ▶ Scalar: $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$.
- ▶ Matrix: $\mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{x}^\top \mathbf{b} + c = \left(\mathbf{x} + \frac{1}{2} \mathbf{A}^{-1} \mathbf{b}\right)^\top \mathbf{A} \left(\mathbf{x} + \frac{1}{2} \mathbf{A}^{-1} \mathbf{b}\right) + \left(c - \frac{1}{4} \mathbf{b}^\top \mathbf{A}^{-1} \mathbf{b}\right)$.

Here $\mathbf{A} = -\frac{1}{2} \mathbf{K}_{\text{post}}^{-1}$, $\mathbf{b} = \tilde{\mathbf{f}}$, $c = 0$ (or an arbitrary const). So

$$\mathbf{x} + \frac{1}{2} \mathbf{A}^{-1} \mathbf{b} = \mathbf{f} + \frac{1}{2} \left(-\frac{1}{2} \mathbf{K}_{\text{post}}^{-1} \right)^{-1} \tilde{\mathbf{f}} = \mathbf{f} - \mathbf{K}_{\text{post}} \tilde{\mathbf{f}},$$

and we drop $c - \frac{1}{4} \mathbf{b}^\top \mathbf{A}^{-1} \mathbf{b}$ (a constant in $\mathbf{f}$, absorbed by $\exp$ and $\propto$).

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With $\mathbf{A} = -\frac{1}{2}\mathbf{K}_{\text{post}}^{-1}$, the exponent becomes a clean quadratic form in $\mathbf{f}$:

$$\begin{aligned} p(\mathbf{f}|\mathbf{y}) &\propto \exp\left(-\frac{1}{2}(\mathbf{f} - \mathbf{K}_{\text{post}}\tilde{\mathbf{f}})^{\top}\mathbf{K}_{\text{post}}^{-1}(\mathbf{f} - \underbrace{\mathbf{K}_{\text{post}}\tilde{\mathbf{f}}}_{:=\mathbf{f}_{\text{post}}})\right) \\ &\propto \exp\left(-\frac{1}{2}(\mathbf{f} - \mathbf{f}_{\text{post}})^{\top}\mathbf{K}_{\text{post}}^{-1}(\mathbf{f} - \mathbf{f}_{\text{post}})\right). \end{aligned}$$

This is the kernel of a Gaussian with covariance $\mathbf{K}_{\text{post}}$ and mean $\mathbf{f}_{\text{post}}$.
Hence

$$\mathbf{f}|\mathbf{y} \sim \mathcal{N}(\mathbf{f}_{\text{post}}, \mathbf{K}_{\text{post}}). \quad \square$$

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Exercise 1 (c)

(c) Derive the posterior predictive distribution $y_* | x_*, \mathbf{x}, \mathbf{y}$ for a new sample x_* from the **same** data-generating process.

1 (c): Think Before Solving

Target: the distribution of a *new noisy* target y_* at a test point x_* , given the training data.

- ▶ **The integral is cumbersome.** Marginalizing $\mathbf{f}$ out of $p(y_* | \dots, \mathbf{f}) p(\mathbf{f} | \mathbf{y})$ works but is messy. Is there a shortcut that avoids integrating?
- ▶ **Use the known joint.** Training targets $\mathbf{y}$ and the new y_* are *jointly* Gaussian. What are the four covariance blocks of $(\mathbf{y}, y_*)$? Where does the noise σ^2 enter?
- ▶ **Condition, don't integrate.** The Gaussian conditioning rule turns the joint into $y_* | \mathbf{y}$ directly. What are $\mu_{a|b}$ and $\Sigma_{a|b}$ here?

known joint Gaussian $(\mathbf{y}, y_*) \rightarrow$ apply conditioning rule $\rightarrow$ done (no integral)

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1 (c): Predictive via Gaussian Conditioning

Naïvely: marginalize $\mathbf{f}$ out via $p(y_*|x_*, \mathbf{x}, \mathbf{y}) = \int p(y_*|x_*, \mathbf{x}, \mathbf{y}, \mathbf{f}) p(\mathbf{f}|\mathbf{y}, \mathbf{x}) d\mathbf{f}$ — cumbersome.

Instead, use the **Gaussian conditioning rule**:

If $\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}, \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ab}^\top & \Sigma_{bb} \end{pmatrix}\right)$, then $\mathbf{a} | \mathbf{b} \sim \mathcal{N}(\boldsymbol{\mu}_{a|b}, \Sigma_{a|b})$ with

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \Sigma_{ab} \Sigma_{bb}^{-1} (\mathbf{b} - \boldsymbol{\mu}_b),$$

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab} \Sigma_{bb}^{-1} \Sigma_{ba}.$$

1 (c): Predictive via Gaussian Conditioning

Here we condition y_* on $\mathbf{y}$ (not $\mathbf{f}$), and $\mathbf{y} = \mathbf{f} + \epsilon \Rightarrow$ add σ^2 on the diagonal of the cov. matrix.

The joint distribution of $(\mathbf{y}, y_*)$:

$$\begin{pmatrix} \mathbf{y} \\ y_* \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mathbf{m} \\ m_* \end{pmatrix}, \begin{pmatrix} \mathbf{K} + \sigma^2 \mathbf{I} & \mathbf{K}_* \\ \mathbf{K}_*^\top & \mathbf{K}_{**} + \sigma^2 \end{pmatrix} \right).$$

Conditioning ($\mathbf{a} = y_*$, $\mathbf{b} = \mathbf{y}$) gives a Gaussian $y_* | \mathbf{x}_*, \mathbf{x}, \mathbf{y} \sim \mathcal{N}(\mu_*, \Sigma_*)$ with (here $\mathbf{a} = y_*$ is *scalar*, so $\Sigma_{a|b}$ is the scalar predictive variance)

$$\begin{aligned} \mu_* &= m_* + \mathbf{K}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} (\mathbf{y} - \mathbf{m}), \\ \Sigma_* &= \mathbf{K}_{**} + \sigma^2 - \mathbf{K}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{K}_*. \quad \square \end{aligned}$$

NB: the variance keeps the $+\sigma^2$ because y_* is the *observable* target. The latent f_* would drop it: $\text{Var} = \mathbf{K}_{**} - \mathbf{K}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{K}_*$.

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Conditioning ($\mathbf{a} = y_*$, $\mathbf{b} = \mathbf{y}$) gives a Gaussian $y_* | \mathbf{x}_*, \mathbf{x}, \mathbf{y} \sim \mathcal{N}(\mu_*, \Sigma_*)$ with (here $\mathbf{a} = y_*$ is *scalar*, so $\Sigma_{a|b}$ is the scalar predictive variance)

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NB: the variance keeps the $+\sigma^2$ because y_* is the *observable* target. The latent f_* would drop it: $\text{Var} = \mathbf{K}_{**} - \mathbf{K}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{K}_*$.

Roadmap (Ex. 1)

✓ (a) Prior

$\mathbf{f} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$

✓ (b) Posterior

$\mathbf{f} | \mathbf{y}$: complete sq.

► (c) Predictive

$y_* | \dots$: condition

(d) Implement

2×2 by hand + code

1 (c): Predictive via Gaussian Conditioning

Here we condition y_* on $\mathbf{y}$ (not $\mathbf{f}$), and $\mathbf{y} = \mathbf{f} + \epsilon \Rightarrow$ add σ^2 on the diagonal of the cov. matrix.

The joint distribution of $(\mathbf{y}, y_*)$:

$$\begin{pmatrix} \mathbf{y} \\ y_* \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mathbf{m} \\ m_* \end{pmatrix}, \begin{pmatrix} \mathbf{K} + \sigma^2 \mathbf{I} & \mathbf{K}_* \\ \mathbf{K}_*^\top & \mathbf{K}_{**} + \sigma^2 \end{pmatrix} \right).$$

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Exercise 1 (d)

(d) Implement the GP with squared exponential (SE) kernel, zero mean function and $\ell = 1$ from scratch for $n = 2$ observations $(\mathbf{y}, \mathbf{x})$. Do this as efficiently as possible by explicitly calculating all expensive computations **by hand**. Do the same for the posterior predictive distribution of y_* . Test your implementation using simulated data.

1 (d): Think Before Solving

Target: a from-scratch GP for $n = 2$ where the "expensive" step is done analytically.

- ▶ **What is expensive?** The predictive formula needs $(\mathbf{K} + \sigma^2 \mathbf{I})^{-1}$. For $n = 2$ this is just a 2×2 inverse — can you write it in closed form?
- ▶ **Few distinct entries.** With the SE kernel and $\ell = 1$, $k(x, x) = 1$ and off-diagonals share one value ρ . How many numbers actually determine $\mathbf{K}$?
- ▶ **Training.** With zero mean and ℓ fixed, the only free parameter is σ^2 . How do you pick it from the data (which Gaussian do you maximize)?

2×2 inverse by hand $\rightarrow$ plug into part (c) $\rightarrow$ tune σ^2 by marginal likelihood

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1 (d): Kernel Matrix and Its Inverse by Hand

SE kernel, $\ell = 1$: $k(x, x') = \exp(-\frac{1}{2}(x - x')^2)$, so $k(x, x) = 1$. For $n = 2$ set $\rho := \exp(-\frac{1}{2}(x^{(1)} - x^{(2)})^2) \in (0, 1]$:

$$\mathbf{K} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \mathbf{K}^{-1} = \frac{1}{1 - \rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}.$$

Noisy version $\mathbf{K}_y := \mathbf{K} + \sigma^2 \mathbf{I}$ (the matrix the predictive actually inverts):

$$\mathbf{K}_y = \begin{pmatrix} 1 + \sigma^2 & \rho \\ \rho & 1 + \sigma^2 \end{pmatrix}, \quad \mathbf{K}_y^{-1} = \frac{1}{(1 + \sigma^2)^2 - \rho^2} \begin{pmatrix} 1 + \sigma^2 & -\rho \\ -\rho & 1 + \sigma^2 \end{pmatrix}.$$

($\rho^2 = \exp(-(x^{(1)} - x^{(2)})^2)$.) This closed-form 2×2 inverse is the "expensive computation by hand."



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1 (d): Posterior Predictive at x_* (Zero Mean)

With zero mean ($\mathbf{m} = \mathbf{0}$, $m_* = 0$) plug $\mathbf{K}_y^{-1}$ into the part-(c) formula. For a test point x_* ,

$$\mathbf{K}_* = (k(x_*, x^{(1)}), k(x_*, x^{(2)}))^\top, \quad \mathbf{K}_{**} = k(x_*, x_*) = 1.$$

$$\text{mean} = \mathbf{K}_*^\top \mathbf{K}_y^{-1} \mathbf{y},$$

$$\text{var} = \underbrace{\mathbf{K}_{**} + \sigma^2}_{= 1 + \sigma^2} - \mathbf{K}_*^\top \mathbf{K}_y^{-1} \mathbf{K}_*.$$

Training (σ^2). Zero mean, ℓ fixed $\Rightarrow$ the only free parameter is σ^2 . Maximize the log marginal likelihood of $\mathbf{y} \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_y)$. Using $\partial \mathbf{K}_y / \partial \sigma^2 = \mathbf{I}$, set the derivative to zero:

$$\text{tr}(\mathbf{K}_y^{-1} \mathbf{y} \mathbf{y}^\top \mathbf{K}_y^{-1} - \mathbf{K}_y^{-1}) = 0,$$

and solve for σ^2 numerically by root-finding. □

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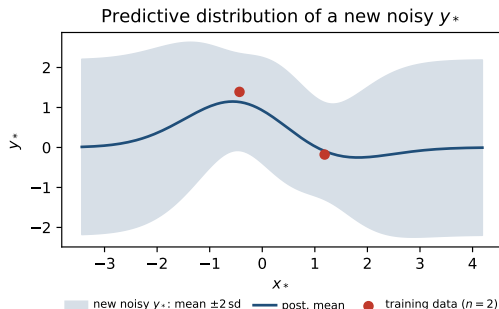
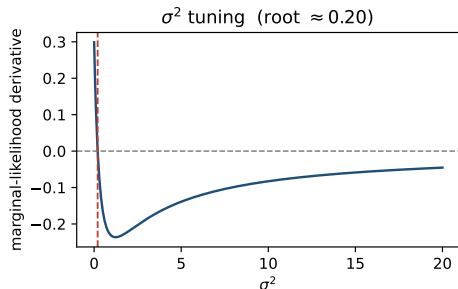
1 (d): Implementation and Test (Python)

Pure NumPy/SciPy (no GP library); the prediction uses the part-(c) result.

```
import numpy as np; from scipy.optimize import brentq
rng = np.random.default_rng(190); n = 2; sigma = 1.0
def kern(x, y=None):                                # SE kernel, l = 1
    d = np.subtract.outer(np.ravel(x), np.ravel(x if y is None else y))
    return np.exp(-0.5 * d**2)                       # k(x,y) = exp(-0.5 (x-y)^2)
# simulate noisy targets y ~ N(0, K + sigma^2 I)
x = rng.standard_normal(n); Ky = kern(x) + sigma**2 * np.eye(n)
y = rng.multivariate_normal(np.zeros(n), Ky).reshape(-1, 1)
# tune sigma^2 by marginal likelihood: solve tr(Kyi yy^T Kyi - Kyi) = 0
def root(s2):
    Kyi = np.linalg.inv(kern(x) + s2 * np.eye(n))
    return 0.5 * np.trace(Kyi @ y @ y.T @ Kyi - Kyi)
s2 = brentq(root, 1e-3, 20)
# posterior predictive at x* = 0 (part (c))
Kyi = np.linalg.inv(kern(x) + s2 * np.eye(n))
Ks = kern(x, [0.0]); Kss = 1.0
mean_pred = Ks.T @ Kyi @ y
var_pred = Kss + s2 - Ks.T @ Kyi @ Ks               # +s2: observable y*
```

1 (d): Output — σ^2 Fit and Posterior Predictive

Simulated $n = 2$ data: marginal-likelihood tuning gives $\sigma^2 \approx 0.20$ (left), and the right panel shows the predictive distribution of a new noisy target y_* .



Question: Why is the predictive standard deviation nonzero at the training points, and why do three red points appear although $n = 2$?

Explanation: Zero uncertainty at training points applies only to the latent GP with noiseless observations; here the band predicts a new noisy y_* , and the apparent third red point was only the legend marker.

Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
(a)	Prior $\mathbf{f} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$, $\mathbf{K}_{ij} = k(x^{(i)}, x^{(j)})$.	A GP restricted to finitely many points is a finite-dimensional Gaussian.
(b)	Posterior $\mathbf{f} \mathbf{y} \sim \mathcal{N}(\mathbf{f}_{\text{post}}, \mathbf{K}_{\text{post}})$ via Bayes + completing the square: $\mathbf{K}_{\text{post}}^{-1} = (\sigma^2 \mathbf{I})^{-1} + \mathbf{K}^{-1}$, $\mathbf{f}_{\text{post}} = \mathbf{K}_{\text{post}} \tilde{\mathbf{f}}$.	Bayesian posterior; Gaussian $\times$ Gaussian stays Gaussian.
(c)	Predictive $y_* x_*, \mathbf{x}, \mathbf{y}$ by conditioning the noisy joint $(\mathbf{y}, y_*)$; variance keeps $+\sigma^2$ (observable y_*).	Gaussian conditioning rule; latent f_* vs. observable y_* .
(d)	$n = 2$ SE kernel: closed-form 2×2 inverse $\mathbf{K}_y^{-1}$, plug into (c); tune σ^2 by marginal likelihood (root-finding).	Efficient GP regression; marginal-likelihood training.